

Download TCGA data using TCGAbiolinks

2025-06-30

Set CRAN mirror explicitly

```
options(repos = c(CRAN = "https://cloud.r-project.org"))
```

```
library(TCGAbiolinks)
```

```
## Warning: package 'TCGAbiolinks' was built under R version 4.3.2
```

```
library(SummarizedExperiment)
```

```
## Warning: package 'SummarizedExperiment' was built under R version 4.3.2
```

```
## Loading required package: MatrixGenerics
```

```
## Warning: package 'MatrixGenerics' was built under R version 4.3.3
```

```
## Loading required package: matrixStats
```

```
## Warning: package 'matrixStats' was built under R version 4.3.3
```

```
##
```

```
## Attaching package: 'MatrixGenerics'
```

```
## The following objects are masked from 'package:matrixStats':
```

```
##
```

```
## colAlls, colAnyNAs, colAnys, colAveragesPerRowSet, colCollapse,
## colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
## colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
## colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
## colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
## colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
## colWeightedMeans, colWeightedMedians, colWeightedSds,
## colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAveragesPerColSet,
## rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
## rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
## rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
## rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
## rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
## rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
## rowWeightedSds, rowWeightedVars
```

```

## Loading required package: GenomicRanges

## Warning: package 'GenomicRanges' was built under R version 4.3.3

## Loading required package: stats4

## Loading required package: BiocGenerics

## Warning: package 'BiocGenerics' was built under R version 4.3.2

##
## Attaching package: 'BiocGenerics'

## The following objects are masked from 'package:stats':
##
##     IQR, mad, sd, var, xtabs

## The following objects are masked from 'package:base':
##
##     anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##     colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##     get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
##     match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
##     Position, rank, rbind, Reduce, rownames, sapply, setdiff, sort,
##     table, tapply, union, unique, unsplit, which.max, which.min

## Loading required package: S4Vectors

## Warning: package 'S4Vectors' was built under R version 4.3.3

##
## Attaching package: 'S4Vectors'

## The following object is masked from 'package:utils':
##
##     findMatches

## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname

## Loading required package: IRanges

## Warning: package 'IRanges' was built under R version 4.3.3

## Loading required package: GenomeInfoDb

## Warning: package 'GenomeInfoDb' was built under R version 4.3.2

## Loading required package: Biobase

```

```
## Warning: package 'Biobase' was built under R version 4.3.3

## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase")', and for packages 'citation("pkgname)".

##
## Attaching package: 'Biobase'

## The following object is masked from 'package:MatrixGenerics':
##
##     rowMedians

## The following objects are masked from 'package:matrixStats':
##
##     anyMissing, rowMedians
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.3.3

##
## Attaching package: 'dplyr'

## The following object is masked from 'package:Biobase':
##
##     combine

## The following objects are masked from 'package:GenomicRanges':
##
##     intersect, setdiff, union

## The following object is masked from 'package:GenomeInfoDb':
##
##     intersect

## The following objects are masked from 'package:IRanges':
##
##     collapse, desc, intersect, setdiff, slice, union

## The following objects are masked from 'package:S4Vectors':
##
##     first, intersect, rename, setdiff, setequal, union

## The following objects are masked from 'package:BiocGenerics':
##
##     combine, intersect, setdiff, union
```

```
## The following object is masked from 'package:matrixStats':
##
##     count

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(here)
```

```
## Warning: package 'here' was built under R version 4.3.3
```

```
## here() starts at /mnt/beegfs/mnt/work/bioinfo/home/chu25
```

Download TCGA data using TCGAbiolinks.

```
# Load TCGA data (example for LUAD - Lung Adenocarcinoma)
#LUAD_query <- GDCquery(project = "TCGA-LUAD",
#                        data.category = "Transcriptome Profiling",
#                        data.type = "Gene Expression Quantification",
#                        workflow.type = "STAR - Counts")

#GDCdownload(LUAD_query)

# this returns a summarizedExperiment object
#TCGA_LUAD_data <- GDCprepare(LUAD_query)

#saveRDS(TCGA_LUAD_data, "/home/chu25/TCGA/TCGA_LUAD_SummarizedExperiment.rds")
```

Read in the saved RDS file:

```
TCGA_LUAD_data<- readRDS("/home/chu25/TCGA/TCGA_LUAD_SummarizedExperiment.rds")

## there are many metadata for each sample
colData(TCGA_LUAD_data) %>%
  colnames() %>%
  tail()

## [1] "paper_chromosome.affected.by.chromothripsis"
## [2] "paper_iCluster.Group"
## [3] "paper_CIMP.methylation.signature."
## [4] "paper_MTOR.mechanism.of.mTOR.pathway.activation"
## [5] "paper_Ploidy.ABSOLUTE.calls"
## [6] "paper_Purity.ABSOLUTE.calls"
```

```
# CIMP methylation subtypes
colData(TCGA_LUAD_data) %>%
  as.data.frame() %>%
  janitor::tabyl(~paper_CIMP.methylation.signature.)
```

```
## paper_CIMP.methylation.signature.    n    percent valid_percent
##                               high    37 0.06166667    0.1957672
##                               intermediate 76 0.12666667    0.4021164
##                               low    76 0.12666667    0.4021164
##                               <NA> 411 0.68500000         NA
```

```
# save the raw counts matrix
TCGA_LUAD_mat<- assay(TCGA_LUAD_data)
```

```
TCGA_LUAD_mat[1:5, 1:5]
```

```
##                               TCGA-44-6147-01B-06R-A277-07 TCGA-44-6147-01A-11R-1755-07
## ENSG000000000003.15                                1002                                2834
## ENSG000000000005.6                                  10                                  2
## ENSG000000000419.13                                284                                1307
## ENSG000000000457.14                                549                                715
## ENSG000000000460.17                                497                                285
##                               TCGA-44-7661-01A-11R-2066-07 TCGA-05-4396-01A-21R-1858-07
## ENSG000000000003.15                                2138                                1107
## ENSG000000000005.6                                  4                                  0
## ENSG000000000419.13                                2179                                776
## ENSG000000000457.14                                830                                495
## ENSG000000000460.17                                610                                115
##                               TCGA-38-4629-01A-02R-1206-07
## ENSG000000000003.15                                7434
## ENSG000000000005.6                                  0
## ENSG000000000419.13                                2968
## ENSG000000000457.14                                555
## ENSG000000000460.17                                643
```

```
dim(TCGA_LUAD_mat)
```

```
## [1] 60660    600
```

Download the TCGA LUSC data

```
#LUSC_query <- GDCquery(project = "TCGA-LUSC",
#                         data.category = "Transcriptome Profiling",
#                         data.type = "Gene Expression Quantification",
#                         workflow.type = "STAR - Counts")

#GDCdownload(LUSC_query)

#TCGA_LUSC_data <- GDCprepare(LUSC_query)

#saveRDS(TCGA_LUSC_data, "/home/chu25/TCGA/TCGA_LUSC_SummarizedExperiment.rds")
```

read in the saved RDS

```
TCGA_LUSC_data<- readRDS("/home/chu25/TCGA/TCGA_LUSC_SummarizedExperiment.rds")
```

```
# different transcription subtypes
colData(TCGA_LUSC_data) %>%
  as.data.frame() %>%
  janitor::tabyl(paper_Expression.Subtype)
```

```
## paper_Expression.Subtype    n    percent valid_percent
##                basal    45 0.08007117      0.2432432
##                classical    67 0.11921708      0.3621622
##                primitive    27 0.04804270      0.1459459
##                secretory    46 0.08185053      0.2486486
##                <NA>    377 0.67081851           NA
```

```
TCGA_LUSC_mat<- assay(TCGA_LUSC_data)
```

convert ENSEMBL id to gene symbols

```
library(org.Hs.eg.db)
```

```
## Loading required package: AnnotationDbi
```

```
## Warning: package 'AnnotationDbi' was built under R version 4.3.2
```

```
##
```

```
## Attaching package: 'AnnotationDbi'
```

```
## The following object is masked from 'package:dplyr':
```

```
##
```

```
##      select
```

```
##
```

```
# remove the version number in the end of the ENSEMBL id
TCGA_LUAD_genes<- rownames(TCGA_LUAD_mat) %>%
  tibble::enframe() %>%
  mutate(ENSEMBL =stringr::str_replace(value, "\\.[0-9]+", ""))

head(TCGA_LUAD_genes)
```

```
## # A tibble: 6 x 3
```

```
##   name value      ENSEMBL
##   <int> <chr>      <chr>
## 1     1 ENSG00000000003.15 ENSG00000000003
## 2     2 ENSG00000000005.6  ENSG00000000005
## 3     3 ENSG00000000419.13 ENSG00000000419
## 4     4 ENSG00000000457.14 ENSG00000000457
## 5     5 ENSG00000000460.17 ENSG00000000460
## 6     6 ENSG00000000938.13 ENSG00000000938
```

```
clusterProfiler::bitr(TCGA_LUAD_genes$ENSEMBL,
                      fromType = "ENSEMBL",
                      toType = "SYMBOL",
                      OrgDb = org.Hs.eg.db) %>%
  janitor::get_dupes(SYMBOL) %>%
  head()
```

```
##
```

```
## 'select()' returned 1:many mapping between keys and columns
```

```
## Warning in clusterProfiler::bitr(TCGA_LUAD_genes$ENSEMBL, fromType = "ENSEMBL",
## : 40.28% of input gene IDs are fail to map...
```

```
##          SYMBOL dupe_count      ENSEMBL
## 1 LOC124905422          16 ENSG00000199270
## 2 LOC124905422          16 ENSG00000199334
## 3 LOC124905422          16 ENSG00000199337
## 4 LOC124905422          16 ENSG00000199352
## 5 LOC124905422          16 ENSG00000199396
## 6 LOC124905422          16 ENSG00000199910
```

```
# just keep one of them (simple solution)
```

```
TCGA_LUAD_gene_map <- clusterProfiler::bitr(TCGA_LUAD_genes$ENSEMBL,
                                             fromType = "ENSEMBL",
                                             toType = "SYMBOL",
                                             OrgDb = org.Hs.eg.db) %>%
  distinct(SYMBOL, .keep_all = TRUE)
```

```
## 'select()' returned 1:many mapping between keys and columns
```

```
## Warning in clusterProfiler::bitr(TCGA_LUAD_genes$ENSEMBL, fromType = "ENSEMBL",
## : 40.28% of input gene IDs are fail to map...
```

```
TCGA_LUAD_gene_map <- TCGA_LUAD_gene_map %>%
  left_join(TCGA_LUAD_genes)
```

```
## Joining with 'by = join_by(ENSEMBL)'
```

```
head(TCGA_LUAD_gene_map)
```

```
##          ENSEMBL SYMBOL name      value
## 1 ENSG00000000003 TSPAN6    1 ENSG00000000003.15
## 2 ENSG00000000005 TNMD      2 ENSG00000000005.6
## 3 ENSG000000000419 DPM1     3 ENSG000000000419.13
## 4 ENSG000000000457 SCYL3    4 ENSG000000000457.14
## 5 ENSG000000000460 FIRRM    5 ENSG000000000460.17
## 6 ENSG000000000938 FGR      6 ENSG000000000938.13
```

```
TCGA_LUSC_mat<- TCGA_LUSC_mat[TCGA_LUAD_gene_map$value, ]
row.names(TCGA_LUSC_mat)<- TCGA_LUAD_gene_map$SYMBOL
TCGA_LUSC_mat[1:5, 1:5]
```

```
##          TCGA-56-7221-01A-11R-2045-07 TCGA-43-7657-01A-31R-2125-07
## TSPAN6          6066          2329
## TNMD            9            0
## DPM1           3344          2533
## SCYL3           856          496
## FIRRM          1072          434
##          TCGA-43-7657-11A-01R-2125-07 TCGA-60-2695-01A-01R-0851-07
## TSPAN6          2459          4271
## TNMD            0            1
## DPM1           1734          2692
## SCYL3           795          2494
## FIRRM          206          2850
##          TCGA-60-2710-01A-01R-0851-07
## TSPAN6          2550
## TNMD            2
## DPM1           1730
## SCYL3           726
## FIRRM          712
```

```
TCGA_LUAD_mat<- TCGA_LUAD_mat[TCGA_LUAD_gene_map$value, ]
row.names(TCGA_LUAD_mat)<- TCGA_LUAD_gene_map$SYMBOL
TCGA_LUAD_mat[1:5, 1:5]
```

```
##          TCGA-44-6147-01B-06R-A277-07 TCGA-44-6147-01A-11R-1755-07
## TSPAN6          1002          2834
## TNMD            10            2
## DPM1           284          1307
## SCYL3           549          715
## FIRRM          497          285
##          TCGA-44-7661-01A-11R-2066-07 TCGA-05-4396-01A-21R-1858-07
## TSPAN6          2138          1107
## TNMD            4            0
## DPM1           2179          776
## SCYL3           830          495
## FIRRM          610          115
##          TCGA-38-4629-01A-02R-1206-07
## TSPAN6          7434
## TNMD            0
## DPM1           2968
## SCYL3           555
## FIRRM          643
```

```
## check the dimensions
dim(TCGA_LUSC_mat)
```

```
## [1] 36816 562
```



```
## check the dimensions
dim(TCGA_LUAD_mat)
```

```
## [1] 36816 600
```

combine TCGA LUSC and LUAD PCA plot use top variable genes.

```
# double check the genes are the same
all.equal(rownames(TCGA_LUAD_mat), rownames(TCGA_LUSC_mat))
```

```
## [1] TRUE
```

```
TCGA_lung_mat<- cbind(TCGA_LUSC_mat, TCGA_LUAD_mat)

TCGA_lung_meta<- data.frame(cancer_type = c(rep( "LUSC", ncol(TCGA_LUSC_mat)),
      rep("LUAD", ncol(TCGA_LUAD_mat))))

dim(TCGA_lung_mat)
```

```
## [1] 36816 1162
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3
```

```
library(ggfortify)
```

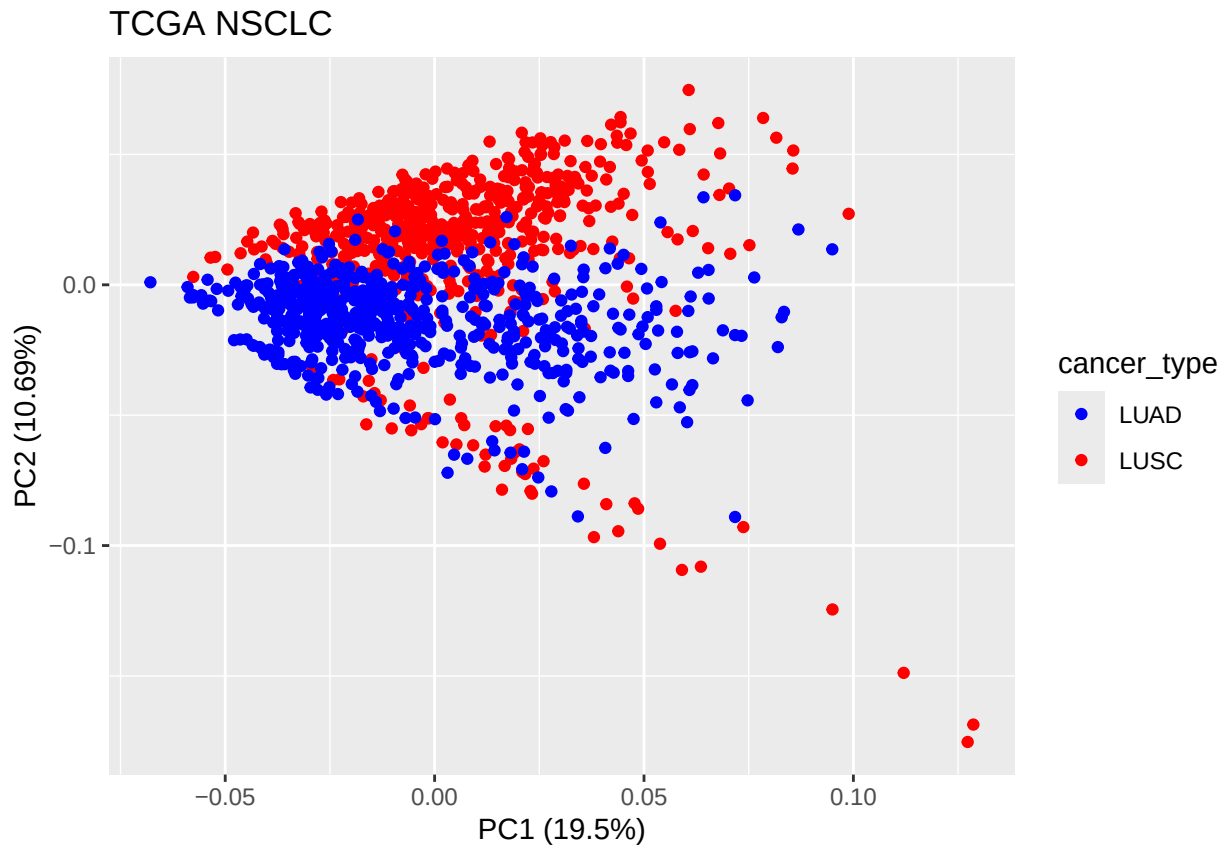
```
## Warning: package 'ggfortify' was built under R version 4.3.3
```

```
# select the top 1000 most variable genes
TCGA_gene_idx<- order(rowVars(TCGA_lung_mat), decreasing = TRUE)[1:1000]

TCGA_lung_mat_sub <- TCGA_lung_mat[TCGA_gene_idx, ]

TCGA_pca_res <- prcomp(t(TCGA_lung_mat_sub), scale. = TRUE)

autoplot(TCGA_pca_res, data = TCGA_lung_meta , color ="cancer_type") +
  scale_color_manual(values = c("blue", "red")) +
  ggtitle("TCGA NSCLC")
```



```
# the PCs
TCGA_pca_res$x[1:5, 1:5]
```

```
##              PC1      PC2      PC3      PC4
## TCGA-56-7221-01A-11R-2045-07 11.099020 15.292867 -3.5023210 -9.426812
## TCGA-43-7657-01A-31R-2125-07 13.523521 14.463327 -4.3607699 -1.723640
## TCGA-43-7657-11A-01R-2125-07  9.102947 -22.676106 -7.0151659 -9.203768
## TCGA-60-2695-01A-01R-0851-07  8.348313  5.639156 -0.4066455  3.399322
## TCGA-60-2710-01A-01R-0851-07 11.635740 10.593525 -14.0096595 -3.221826
##              PC5
## TCGA-56-7221-01A-11R-2045-07  7.941908
## TCGA-43-7657-01A-31R-2125-07 -1.855825
## TCGA-43-7657-11A-01R-2125-07 -2.278144
## TCGA-60-2695-01A-01R-0851-07  2.877935
## TCGA-60-2710-01A-01R-0851-07  6.783877
```

```
seq_depth<- colSums(TCGA_lung_mat)
```

```
# calculate the correlation of first PC and the sequencing depth
# the sign of the PCs are arbitrary, so let's get the absolute number
cor(TCGA_pca_res$x[, 1], seq_depth) %>%
  abs()
```

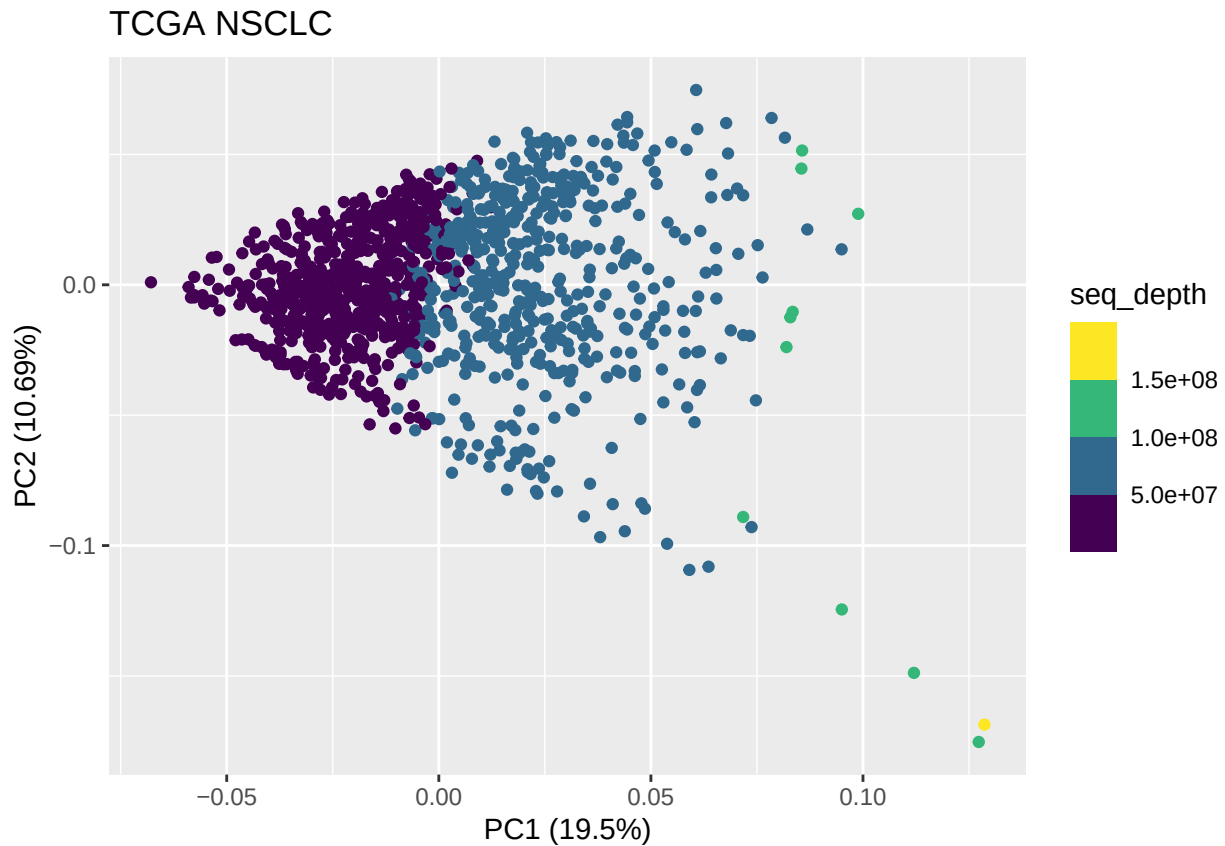
```
## [1] 0.9785658
```

```
# PC2 correlation with sequencing depth
cor(TCGA_pca_res$x[, 2], seq_depth) %>%
  abs()
```

```
## [1] 0.0942689
```

```
TCGA_lung_meta$seq_depth<- seq_depth

autoplot(TCGA_pca_res, data = TCGA_lung_meta , color ="seq_depth") +
  scale_color_viridis_b() +
  ggtitle("TCGA NSCLC")
```



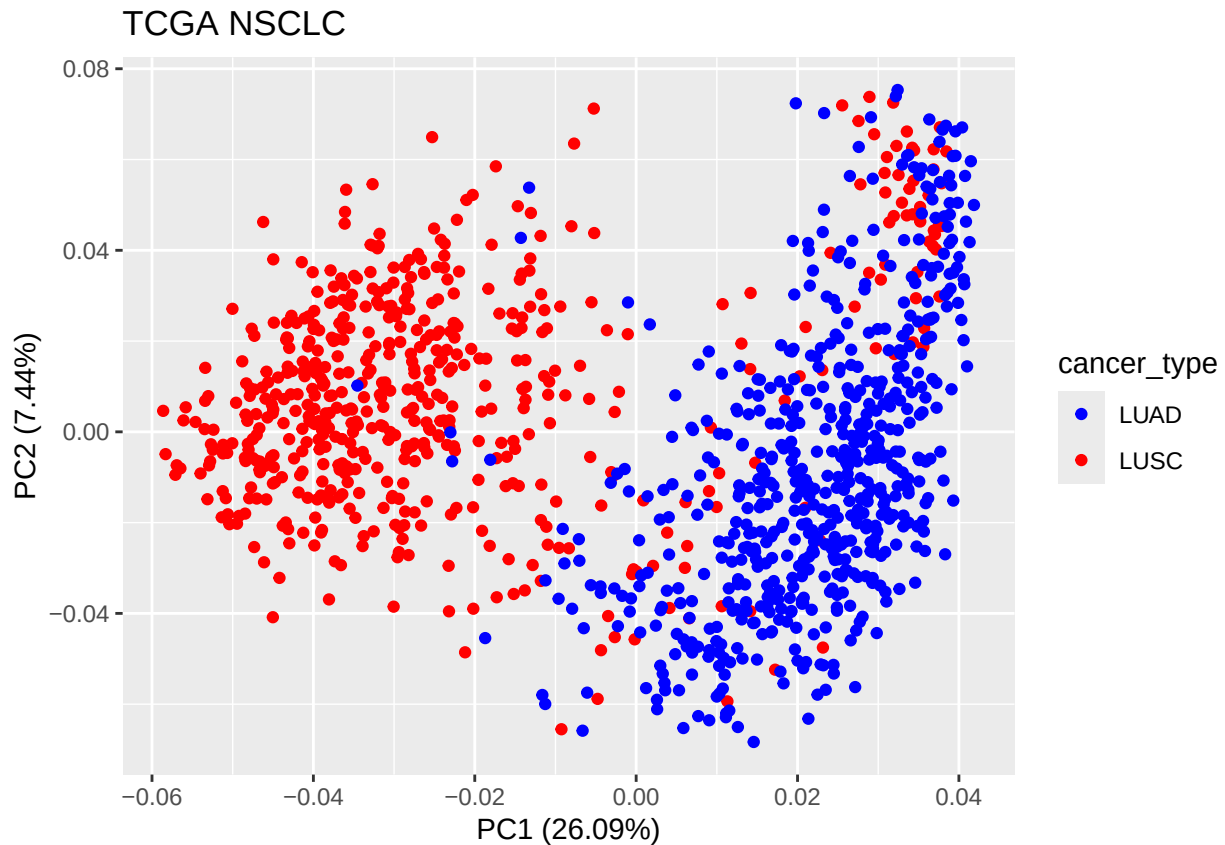
```
# convert the raw counts to log2(cpm+1) using edgeR.
TCGA_lung_cpm<- edgeR::cpm(TCGA_lung_mat, log = TRUE, prior.count = 1)

## top 1000 most variable genes
TCGA_gene_idx2<- order(rowVars(TCGA_lung_cpm), decreasing = TRUE)[1:1000]

TCGA_lung_cpm_sub <- TCGA_lung_cpm[TCGA_gene_idx2, ]

TCGA_pca_res2 <- prcomp(t(TCGA_lung_cpm_sub), scale. = TRUE)

autoplot(TCGA_pca_res2, data = TCGA_lung_meta , color ="cancer_type") +
  scale_color_manual(values = c("blue", "red")) +
  ggtitle("TCGA NSCLC")
```



```
set.seed(123)
library(ComplexHeatmap)
```

```
## Warning: package 'ComplexHeatmap' was built under R version 4.3.2
```

```
## Loading required package: grid
```

```
## =====
## ComplexHeatmap version 2.18.0
## Bioconductor page: http://bioconductor.org/packages/ComplexHeatmap/
## Github page: https://github.com/jokergoo/ComplexHeatmap
## Documentation: http://jokergoo.github.io/ComplexHeatmap-reference
##
## If you use it in published research, please cite either one:
## - Gu, Z. Complex Heatmap Visualization. iMeta 2022.
## - Gu, Z. Complex heatmaps reveal patterns and correlations in multidimensional
##   genomic data. Bioinformatics 2016.
##
##
## The new InteractiveComplexHeatmap package can directly export static
## complex heatmaps into an interactive Shiny app with zero effort. Have a try!
##
## This message can be suppressed by:
## suppressPackageStartupMessages(library(ComplexHeatmap))
## =====
```

```
TCGA_ha<- HeatmapAnnotation(df = TCGA_lung_meta,
  col = list(
    cancer_type = c("LUSC" = "red", "LUAD" = "blue"))
)

Heatmap(t(scale(t(TCGA_lung_cpm_sub))),
  name = "TCGA lung RNA",
  show_column_names = FALSE,
  show_row_names = FALSE,
  show_row_dend = FALSE,
  #column_split = TCGA_lung_meta$cancer_type,
  top_annotation = TCGA_ha,
  row_names_gp = gpar(fontsize = 3),
  border = TRUE,
  row_km = 3
)
```

